Finding Local Extrema

The First and Second Derivative Tests

Friday, November 6, 2026 *Transcribed by Cleo McBride*

Goal. Given a critical number c, determine whether $f(c)$ is a local maximum, a local minimum, or neither.

Theorem 1 (First derivative test). Let c be an interior critical number of f, with f continuous at c. Suppose that on each side of c, sufficiently close to c, f' exists and is either strictly positive throughout or strictly negative throughout. Then

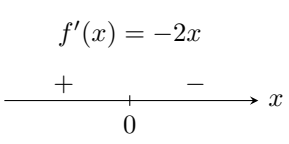
$$f \text{ has a local extremum at } c \iff f' \text{ changes sign at } c.$$

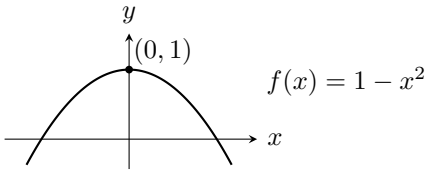
In particular, reading from left to right:

$$\begin{aligned} + \text{ to } - &\implies \text{strict local maximum at } c, \\ - \text{ to } + &\implies \text{strict local minimum at } c. \end{aligned}$$

If the signs agree, f has no local extremum at c.

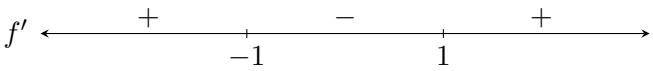
Note. We say that f' *changes sign at* c if, for some $\delta > 0$, f' is positive throughout $(c-\delta, c)$ and negative throughout $(c, c+\delta)$, or vice versa.





Proof (postponed). We will use the following consequence of the Mean Value Theorem. If $f' > 0$ on an interval, then f is increasing there. Similarly, if $f' < 0$, then f is decreasing.

Example 1. Classify the interior critical numbers of $f(x) = 4x^3 - 12x + 12$ on $[-2, 3]$. Its derivative $f'(x) = 12(x-1)(x+1)$ vanishes at -1 and 1.

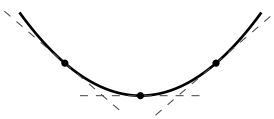


The sign changes from + to − at -1, giving a local maximum of $f(-1) = 20$. At 1, it changes from − to +, giving a local minimum of $f(1) = 4$.

Example 2. For $p(x) = \cos x$, the derivative $p'(x) = -\sin x$ is positive on $(-\pi/2, 0)$ and negative on $(0, \pi/2)$. Hence $p(0) = 1$ is a local maximum.

Concavity and inflection points

Definition 1 (Concavity). On an interval where f is differentiable, its graph is *concave up* if f' is increasing, and *concave down* if f' is decreasing.



Increasing slopes

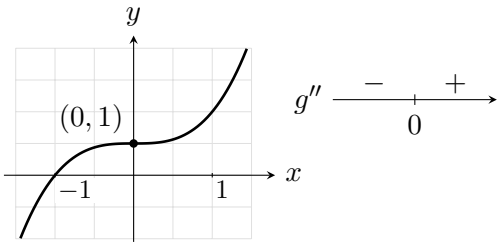


Decreasing slopes

Theorem 2 (Concavity test). If $f'' > 0$ throughout an interval, the graph of f is concave up there. If $f'' < 0$ throughout an interval, it is concave down.

Definition 2 (Inflection point). If f is continuous at c and the graph changes concavity at c, then $(c, f(c))$ is an *inflection point*.

Example 3. Where does $g(x) = x^3 + 1$ change concavity? Since $g'(x) = 3x^2$ and $g''(x) = 6x$, the graph is concave down on $(-\infty, 0)$ and concave up on $(0, \infty)$. The inflection point is $(0, 1)$. The tangent is horizontal there, and g increases through the point.



The second derivative test

Theorem 3 (Second derivative test). Suppose $f'(c) = 0$ and f'' is continuous on an open interval containing c.

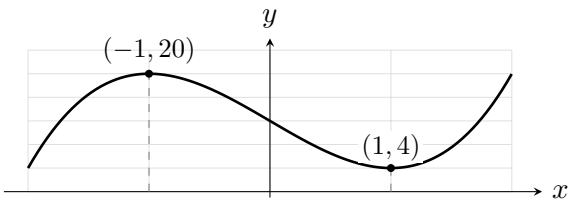
$$f''(c) > 0 \implies f(c) \text{ is a strict local minimum,}$$
$$f''(c) < 0 \implies f(c) \text{ is a strict local maximum.}$$

When $f''(c) = 0$, the test is inconclusive.

Proof. If $f''(c) > 0$, continuity keeps f'' positive near c. Thus f' increases through $f'(c) = 0$, from negative values to positive values. The first derivative test gives a local minimum. When $f''(c) < 0$, f' decreases through zero, giving a local maximum. □

Example 4. For the cubic $f(x) = 4x^3 - 12x + 12$ in Example 1, $f'(-1) = f'(1) = 0$ and $f''(x) = 24x$.

$$f''(-1) = -24 < 0 \implies \text{local maximum at } (-1, 20),$$
$$f''(1) = 24 > 0 \implies \text{local minimum at } (1, 4).$$



Example 5. Use the second derivative test to classify $x = 0$ for $f(x) = x \tan x$. The product rule gives

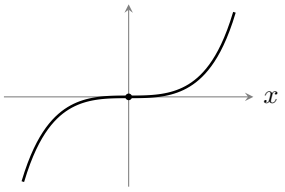
$$f'(x) = \tan x + x \sec^2 x, \qquad f''(x) = 2 \sec^2 x (1 + x \tan x).$$

Since $f'(0) = 0$ and $f''(0) = 2 > 0$, f has a strict local minimum at $(0, 0)$.

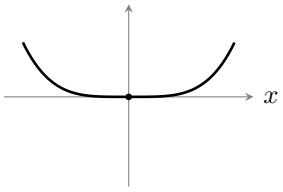
Practice. Now let $h(x) = x^2 \tan x$. Compute $h'(0)$ and $h''(0)$. What does the second derivative test tell you? Is there a local extremum at 0? Justify your answer.

Check. Both derivatives vanish at 0, so the test is inconclusive. Near zero, $x^2 \tan x$ is negative on the left and positive on the right, so $h(0) = 0$ is neither a maximum nor a minimum locally.

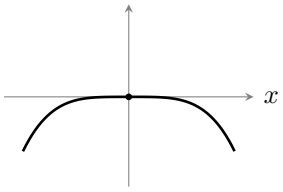
Comparison. All three functions below satisfy $f'(0) = f''(0) = 0$. At 0, x^4 has a local minimum and $-x^4$ has a local maximum.



$y = x^2 \tan x$



$y = x^4$



$y = -x^4$